

PRNGs and Diffie-Hellman Key Exchange

CSE 405 January 2025
Lecture 6

Last Time: Hashes

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- Map arbitrary-length input to fixed-length output
- Output is deterministic and unpredictable
- Security properties
 - One way: Given an output y , it is infeasible to find any input x such that $H(x) = y$.
 - Collision resistant: It is infeasible to find another any pair of inputs $x' \neq x$ such that $H(x) = H(x')$.
- Some hashes are vulnerable to length extension attacks
- Hashes don't provide integrity (unless you can publish the hash securely)

Last Time: MACs

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- Inputs: a secret key and a message
- Output: a tag on the message
- A secure MAC is unforgeable: Even if Mallory can trick Alice into creating MACs for messages that Mallory chooses, Mallory cannot create a valid MAC on a message that she hasn't seen before
- Example: $\text{HMAC}(K, M) = H((K' \oplus \text{opad}) \parallel H((K' \oplus \text{ipad}) \parallel M))$
- MACs do not provide confidentiality

Last Time: Authenticated Encryption

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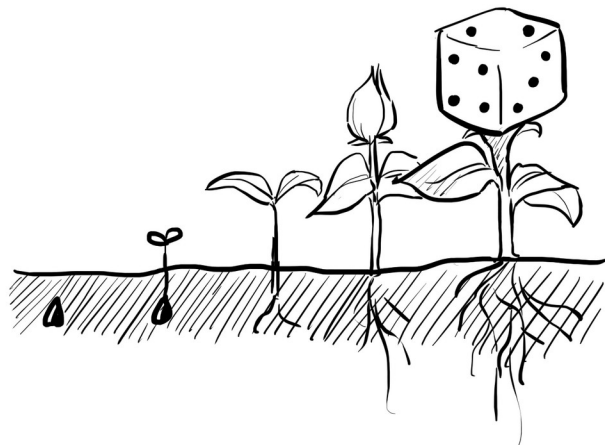
- Authenticated encryption: A scheme that simultaneously guarantees confidentiality and integrity (and authenticity) on a message
- First approach: Combine schemes that provide confidentiality with schemes that provide integrity and authenticity
 - MAC-then-encrypt: $\text{Enc}(K_1, M \parallel \text{MAC}(K_2, M))$
 - Encrypt-then-MAC: $\text{Enc}(K_1, M) \parallel \text{MAC}(K_2, \text{Enc}(K_1, M))$
 - Always use Encrypt-then-MAC because it's more robust to mistakes
- Second approach: Use AEAD encryption modes designed to provide confidentiality, integrity, and authenticity
 - Drawback: Incorrectly using AEAD modes leads to losing *both* confidentiality and integrity/authentication

Today: PRNGs and Diffie-Hellman Key Exchange

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- Symmetric-key encryption schemes need randomness. How do we securely generate random numbers?
- When discussing symmetric-key schemes, we assumed Alice and Bob managed to share a secret key. How can Alice and Bob share a symmetric key over an insecure channel?

Pseudorandom Number Generators (PRNGs)



Textbook Chapter 9

Cryptography Roadmap

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	Symmetric-key	Asymmetric-key
Confidentiality	● One-time pads ● Block ciphers with chaining modes (e.g. AES-CBC)	● RSA encryption ● ElGamal encryption
Integrity, Authentication	● MACs (e.g. HMAC)	● Digital signatures (e.g. RSA signatures)

- Hash functions
- Pseudorandom number generators
- Public key exchange (e.g. Diffie-Hellman)

- Key management (certificates)
- Password management

Randomness

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- Randomness is essential for symmetric-key encryption
 - A random key
 - A random IV/nonce
 - Universally unique identifiers (we'll see this shortly)
 - We'll see more applications later
- If an attacker can predict a random number, things can catastrophically fail
 - Think about NID?
 - Birth Registration Number?
 - How should you assign ID in your 408 project?
- How do we securely generate random numbers?

Entropy

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- In cryptography, “random” usually means “random and unpredictable”
- Scenario
 - You want to generate a secret bitstring that the attacker can't guess
 - You generate random bits by tossing a fair (50-50) coin
 - The outcomes of the fair coin are harder for the attacker to guess
- **Entropy: A measure of uncertainty**
 - In other words, a measure of how unpredictable the outcomes are
 - High entropy = unpredictable outcomes = desirable in cryptography
 - The uniform distribution has the highest entropy (every outcome equally likely, e.g. fair coin toss)
 - Usually measured in bits (so 3 bits of entropy = uniform, random distribution over 8 values)

Breaking Bitcoin Wallets

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- What happens if we use a poor source of entropy?
- Bitcoin users use a randomly-generated private key to access their account (and money)
 - An attacker who learns the key can access the money
 - We'll learn more about Bitcoin later
- An “improvement” [sic] to the algorithm reduced the entropy used to generate the private keys
 - Any private key created with this “improvement” could be brute-forced

Improvements to RNG

committed on Dec 7, 2014 1 parent b0d5639

Showing 1 changed file with 26 additions and 28 deletions.

54 bitcoinjs-lib/src/jsbn/rng.js

```
@@ -8,15 +8,16 @@ var rng_state;
8      var rng_pool;
9      var rng_pptr;
10
11 - // Mix in a 32-bit integer into the pool
12 - function rng_seed_int(x) {
13 -     rng_pool[rng_pptr++] ^= x & 255;
14 -     rng_pool[rng_pptr++] ^= (x >> 8) & 255;
15 -     rng_pool[rng_pptr++] ^= (x >> 16) & 255;
16 -     rng_pool[rng_pptr++] ^= (x >> 24) & 255;
```

XZ Utils Backdoor

CVE-2024-3094 | **CVSS: 10.0**

<https://www.youtube.com/watch?v=F7iLfuci75Y>

True Randomness

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- To generate truly random numbers, we need a physical source of entropy
 - An unpredictable circuit on a CPU
 - Human activity measured at very fine time scales (e.g. the microsecond you pressed a key)
- Unbiased entropy usually requires combining multiple entropy sources
 - Goal: Make the total randomness equal the sum of each source's entropy.
 - Many poor sources + 1 good source = good entropy
- Issues with true randomness
 - It's expensive and slow to generate
 - Physical entropy sources are often biased



Exotic entropy source: Cloudflare has a wall of lava lamps that are recorded by an HD video camera that views the lamps through a rotating prism

[The Lava Lamps That Help Keep The Internet Secure](#)

Pseudorandom Number Generators (PRNGs)

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- True randomness is expensive
- **Pseudorandom number generator (PRNGs)**: An algorithm that uses a little bit of true randomness (seed) to generate a lot of random-looking output
 - Also called **deterministic random bit generators (DRBGs)**
- Usage
 - Generate some expensive true randomness (e.g. noisy circuit on your CPU)
 - Use the true randomness as input to the PRNG
 - Generate random-looking numbers quickly and cheaply with the PRNG
- PRNGs are deterministic: Output is generated according to a set algorithm
 - However, for an attacker who can't see the internal state, the output is *computationally indistinguishable* from true randomness

PRNG: Definition

- A PRNG has three functions:
 - PRNG.Seed(randomness): Initializes the internal state using the entropy
 - Input: Some truly random bits
 - PRNG.Reseed(randomness): Updates the internal state using the existing state and the entropy
 - Input: More truly random bits
 - PRNG.Generate(n): Generate n pseudorandom bits
 - Input: A number n
 - Output: n pseudorandom bits
 - Updates the internal state as needed
- Properties
 - **Correctness**: Deterministic
 - **Efficiency**: Efficient to generate pseudorandom bits
 - **Security**: Indistinguishability from random
 - **Additional security**: Rollback resistance

PRNG: Seeding and Reseeding

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- A PRNG should be seeded with all available sources of entropy
 - Combining many low-entropy sources should result in high-entropy output
 - If one source has 0 entropy (constant), it should not reduce the entropy of the output
- Reseeding is used to add even more entropy as it becomes available
 - Reseeding with 0 entropy should not reduce the entropy of the internal state or output

PRNG: Security

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- Can we design a PRNG that is truly random?
- A PRNG cannot be truly random
 - The output is deterministic given the initial seed
 - If the initial seed is s bits long, there are only 2^s possible output sequences
 - Function: $\{0, 1\}^s \rightarrow \{0, 1\}^{2^s}$ has 2^s inputs and 2^s outputs even though the output space is 2^{2^s}
- A secure PRNG is computationally indistinguishable from random to an attacker
 - Game: Present an attacker with a truly random sequence and a sequence outputted from a secure PRNG
 - An attacker should not be able to determine which is which with probability $> 1/2 + \text{negligible}$
- Equivalent definition: An attacker cannot predict future output of the PRNG

Insecure PRNGs: Breaking Slot Machines

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- What happens if PRNGs are used improperly?
- A casino in St. Louis experienced unusual bad “luck”
 - Suspicious players would hover over the lever and then spin at a specific time to win
- Vulnerability: Slot machines used predictable PRNGs
 - The PRNG output was based on the current time
- Strategy:
 - Use a smartphone to alert you to when to pull the lever for the best chance of winning
- Las Vegas was not affected by the vulnerability
 - Nevada slot machines must follow evaluation standards designed to address this sort of issue

Insecure PRNGs: OpenSSL PRNG bug

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- What happens if we don't use enough entropy?
- Debian OpenSSL CVE-2008-0166
 - Debian: A Linux distribution
 - OpenSSL: A cryptographic library
 - In "cleaning up" OpenSSL (Debian "bug" #363516), the author "fixed" how OpenSSL seeds random numbers
 - The existing code caused Purify and Valgrind to complain about reading uninitialized memory
 - The cleanup caused the PRNG to only be seeded with the process ID
 - There are only 2^{15} (32,768) possible process IDs, so the PRNG only has 15 bits of entropy
- Easy to deduce private keys generated with the PRNG
 - Set the PRNG to every possible starting state and generate a few private/public key pairs
 - See if the matching public key is anywhere on the Internet



PRNG: Rollback Resistance

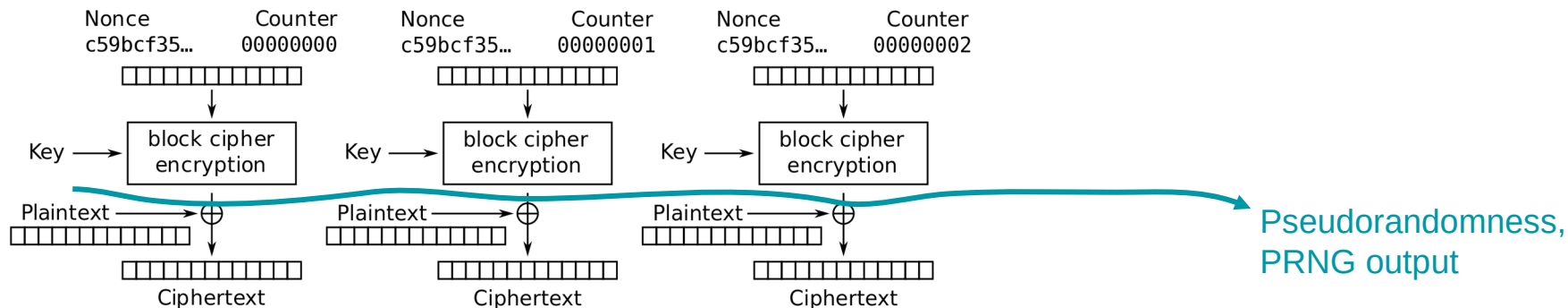
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- **Rollback resistance:** If the attacker learns the internal PRNG state (e.g., by hacking into the machine and dumping its memory), they cannot learn anything about previous states or outputs
 - Game: An attacker knows the current internal state of the PRNG and is given a sequence of truly random bits and a sequence of previous output from the PRNG
 - The attacker cannot determine which is which with probability $> 1/2$
- Rollback resistance is not required in a secure PRNG, but it is a useful property
 - Consider:
 - Alice uses the same PRNG to generate her secret key and the IVs for encryption
 - Mallory compromises the internal state of the PRNG
 - If the PRNG is not rollback resistant, Mallory can derive previous PRNG output... such as the secret key

CTR-DRBG

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- AES in CTR mode generates m pseudo random bits by applying AES $m/128$ times and concatenating the result
- $\text{Generate}(m) = E_k(\text{IV} \parallel 1) \mid E_k(\text{IV} \parallel 2) \mid E_k(\text{IV} \parallel 3) \dots E_k(\text{IV} \parallel \text{ceil}(m/128))$



Counter (CTR) mode encryption

HMAC-DRBG

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- Idea: HMAC output looks unpredictable. Let's use HMAC to build a PRNG!
- HMAC takes two arguments (key and message). Let's keep two values, K (key) and V (value) as internal state
- To generate a block of pseudorandom bits, compute HMAC on the previous block of output.
 - This can be repeated to generate as many pseudorandom bits as needed.

HMAC-DRBG

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Seed(s):

$K := 0$

$V := 0$

Reseed(s)

Initialize internal state



Reseed(s):

$K := \text{HMAC}(K, V \parallel 0x00 \parallel s)$

$V := \text{HMAC}(K, V)$

$K := \text{HMAC}(K, V \parallel 0x01 \parallel s)$

$V := \text{HMAC}(K, V)$

Update internal state with
provided entropy



HMAC-DRBG

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Generate(n):

output = ""

while len(output) < n **do**

$V = \text{HMAC}(K, V)$

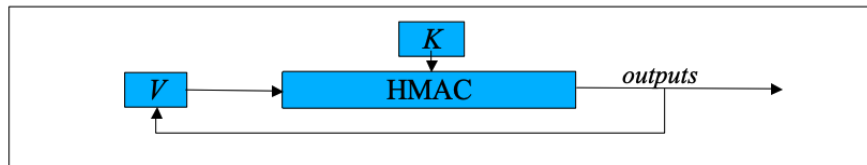
 output = output || V

end while

$K = \text{HMAC}(K, V \parallel 0x00)$

$V = \text{HMAC}(K, V)$

return output[: n]



Call HMAC repeatedly to
generate random-looking output

Update internal state with no
extra entropy

HMAC-DRBG: Security

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- Assuming HMAC is secure, HMAC-DRBG is a secure, rollback-resistant PRNG
 - Secure: If you can distinguish PRNG output from random, then you've distinguished HMAC from random
 - Rollback-resistant: If you can derive old output from the current state, then you've reversed the hash function or HMAC
 - The full proof is out of scope
 - In other words: if you break HMAC-DRBG, you've either broken HMAC or the underlying hash function

Insecure PRNGs: CVE-2019-16303

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- Relevant if you wrote an app in JHipster before 2019
- Password reset functions
 - When you forget your password, receive an email with a special link to reset your password
 - The special link should contain a randomly-generated code (so attackers can't make their own link)
- Vulnerability: Bad PRNG
 - You can figure out the PRNG's internal state from the reset link
 - Request password reset links for other people's accounts
 - Predict the “random” reset link and take over any account you want!

Insecure PRNGs: Rust Rand_Core

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- A Rust library has an interface for “secure” random number generators... but it isn’t actually secure!
- Example: ChaCha8Rng
 - A stream cipher PRNG
 - No reseed function: no way of adding extra entropy after the initial seed
 - Seed only takes 32 bits: no way to combine entropy
 - No rollback resistance
- None of the “secure” RNGs are cryptographically secure
 - None have a reseed function to add extra entropy
 - None take arbitrarily long seeds
- **Takeaway:** Always make sure you use a secure PRNG
 - Consider human factors? Use fail-safe defaults?

Application: Universally Unique Identifiers (UUIDs)

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- Scenario
 - You have a set of objects (e.g. files)
 - You need to assign a unique name to every object
 - Every name must be unique and unpredictable
- Solution: Choose a random value
 - If you use enough randomness, the probability of generating the same random value twice are astronomically small (basically 0)
 - It's more likely that your hardware will fail or that a cosmic ray will flip a bit in your machine's memory than it is that you'll experience a UUID collision.
- Universally Unique Identifiers (UUIDs)
 - 128-bit unique values
 - To generate a new UUID, seed a secure PRNG properly, and generate a random value
 - Often written in hexadecimal: **00112233-4455-6677-8899-aabbccddeeff**
 - You'll work with UUIDs in 408 Project

Stream Ciphers

Textbook Chapter 9.5

Stream Ciphers

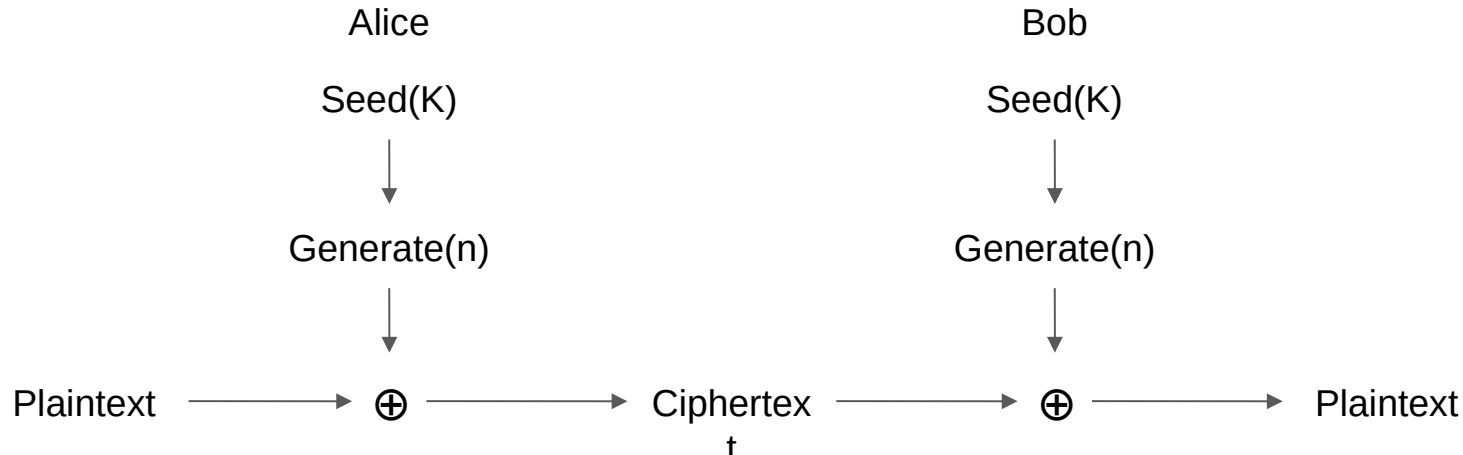
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- Another way to construct symmetric key encryption schemes
- Idea
 - A secure PRNG produces output that looks indistinguishable from random
 - An attacker who can't see the internal PRNG state can't learn any output
 - What if we used PRNG output as the key to a one-time pad?
- **Stream cipher:** A symmetric encryption algorithm that uses pseudorandom bits as the key to a one-time pad

Stream Ciphers

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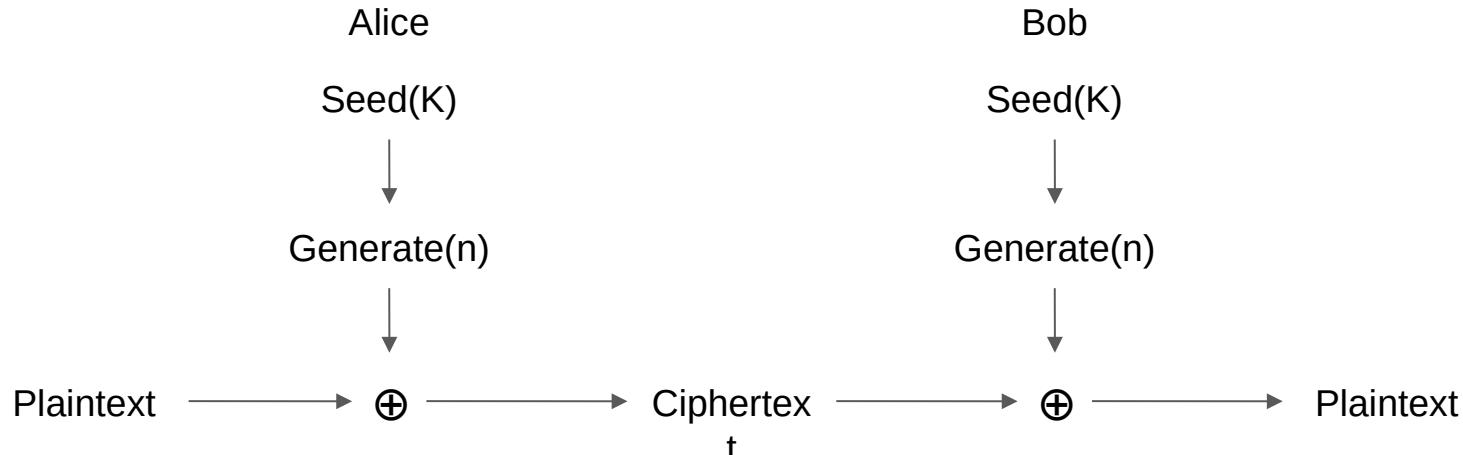
- Protocol: Alice and Bob both seed a secure PRNG with their symmetric secret key, and then use the output as the key for a one-time pad



Stream Ciphers: Encrypting Multiple Messages

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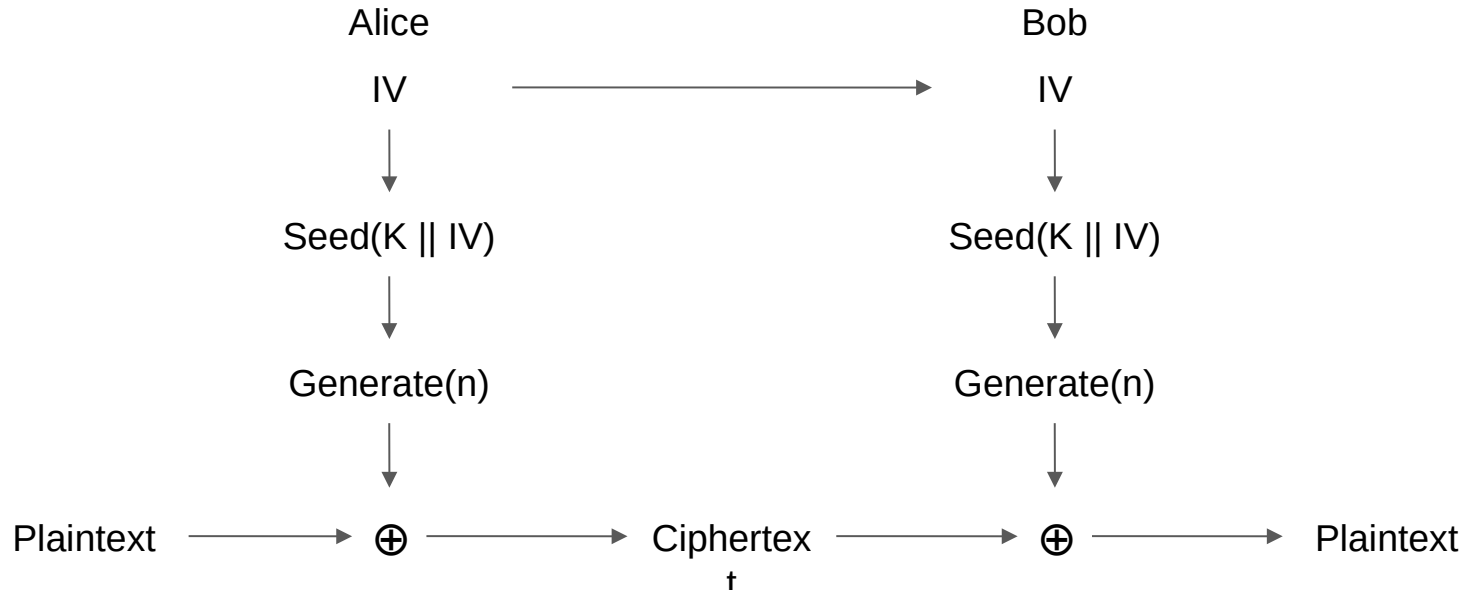
- Recall: One-time pads are insecure when the key is reused. How do we encrypt multiple messages without key reuse?



Stream Ciphers: Encrypting Multiple Messages

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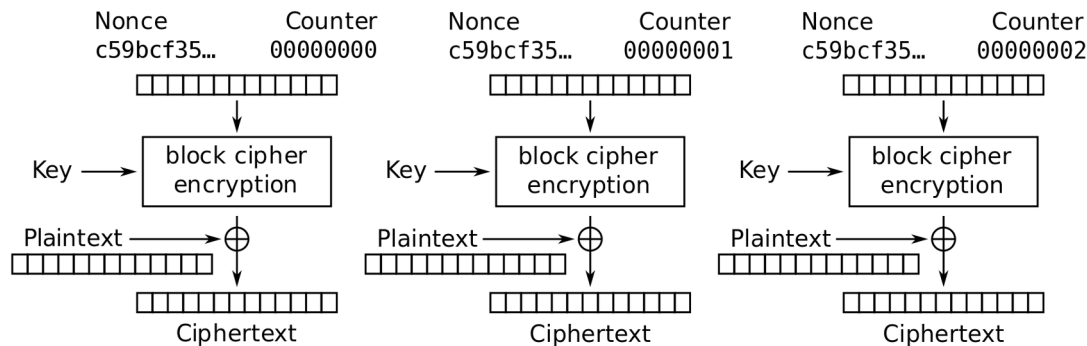
- Solution: For each message, seed the PRNG with the key and a random IV, concatenated. Send the IV with the ciphertext



Stream Ciphers: AES-CTR

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- If you squint carefully, AES-CTR is a type of stream cipher
- Output of the block cipher is pseudorandom and used as a one-time pad



Counter (CTR) mode encryption

Stream Ciphers: Security

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- Stream ciphers are IND-CPA secure, assuming the pseudorandom output is secure
- In some stream ciphers, security is compromised if too much plaintext is encrypted
 - Example: In AES-CTR, if you encrypt so many blocks that the counter wraps around, you'll start reusing keys
 - In practice, if the key is n bits long, usually stop after $2^{n/2}$ bits of output
 - Example: In AES-CTR with 128-bit counters, stop after 2^{64} blocks of output

Stream Ciphers Support Streaming

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- Stream ciphers can continually process new data as it arrives
 - Only need to maintain internal state of the PRNG
 - Keep generating more PRNG output as more input arrives

Stream Ciphers Support Seeking

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- Suppose you received a 1 GB ciphertext (encryption of a 1 GB message) and you only wanted to decrypt the last 128 bytes
- Benefit of some stream ciphers: You can decrypt one part of the ciphertext without decrypting the entire ciphertext
 - Example: In AES-CTR, to decrypt only block i , compute $E_K(\text{nonce} || i)$ and XOR with the i th block of ciphertext
 - Example: ChaCha20 (another stream cipher) lets you decrypt arbitrary parts of ciphertext
 - With HMAC-DRBG, you have to generate all the PRNG output up until the block you want to decrypt

Next: Diffie-Hellman Key Exchange

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- When discussing symmetric-key schemes, we assumed Alice and Bob managed to share a secret key. How can Alice and Bob share a symmetric key over an insecure channel?

Diffie-Hellman Key Exchange

Cryptography Roadmap

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	Symmetric-key	Asymmetric-key
Confidentiality	● One-time pads ● Block ciphers with chaining modes (e.g. AES-CBC)	● RSA encryption ● ElGamal encryption
Integrity, Authentication	● MACs (e.g. HMAC)	● Digital signatures (e.g. RSA signatures)

- Hash functions
- Pseudorandom number generators
- Public key exchange (e.g. Diffie-Hellman)

- Key management (certificates)
- Password management

Diffie-Hellman Key Exchange

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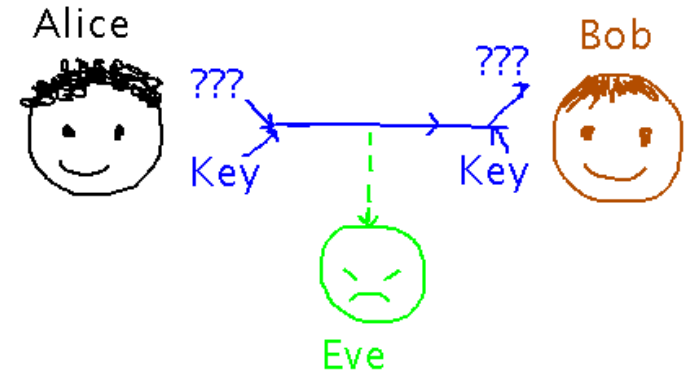
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Sharing Secret Keys

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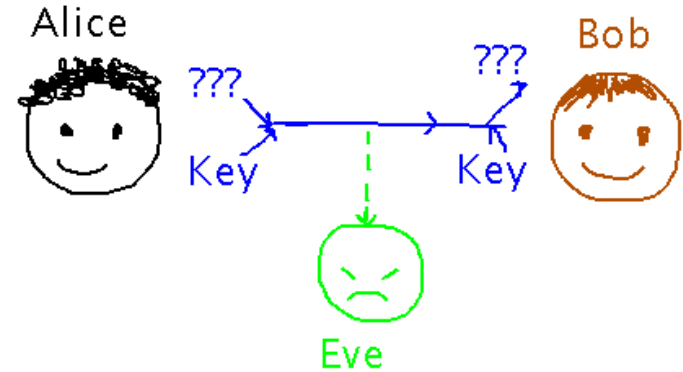
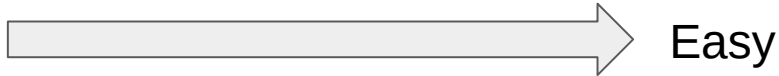
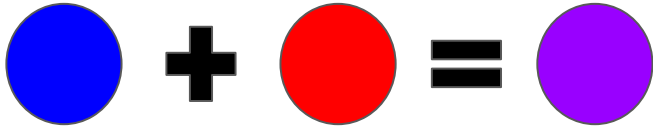
- Eve is always listening
- How would Alice and Bob share a secret key through an insecure channel?



Mixing Colors!

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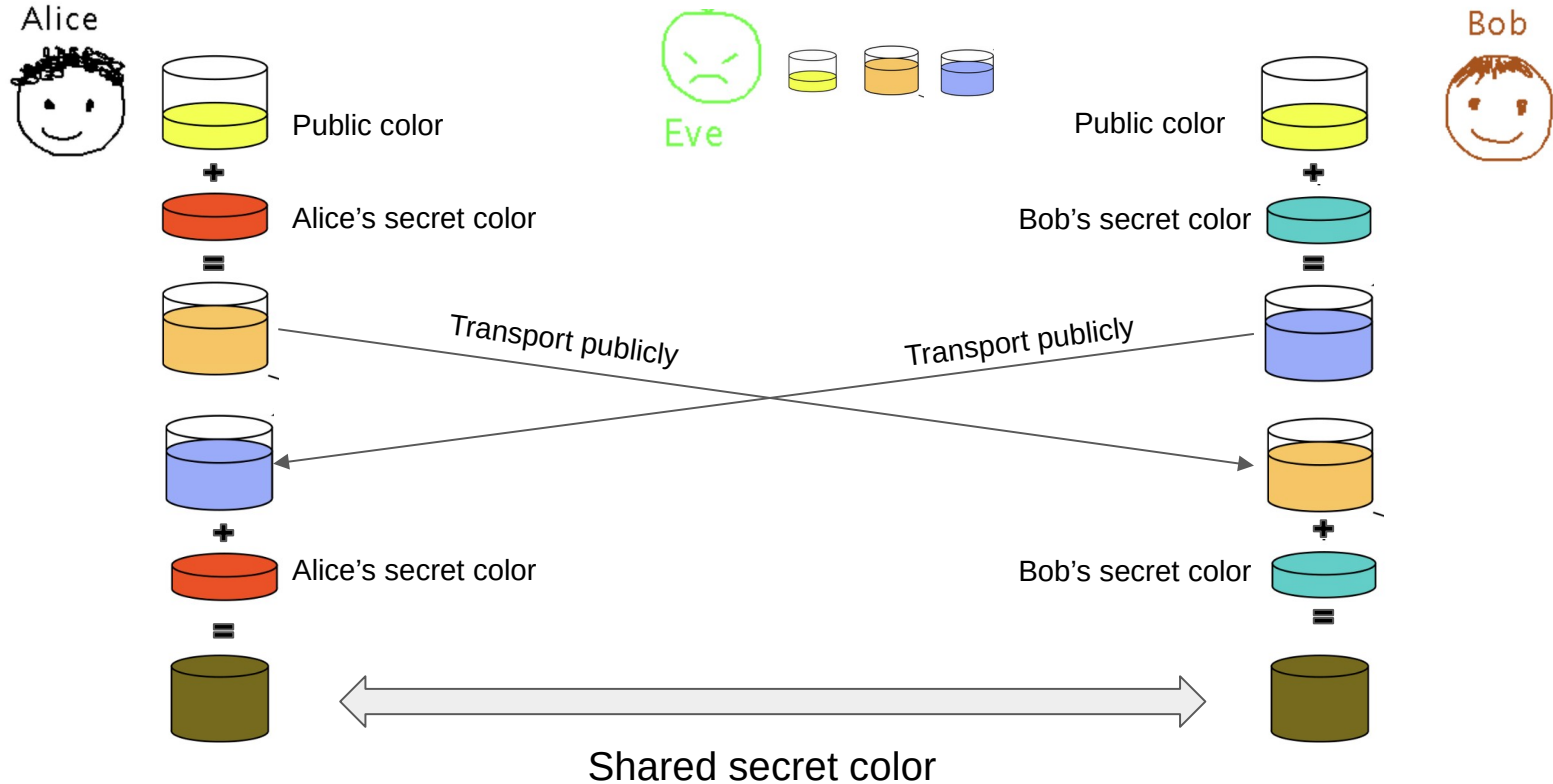
- Suppose the key is a secret color.
- Mixing colors is a one-way function



Mathematical equivalent of color mixing: Discrete logarithm problem

Secure Color Sharing

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Secure Color Sharing

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- Suppose Alice and Bob want a secret *paint color*, but Eve can see paint colors sent between Alice and Bob
- Alice generates a secret color **amber A**, and Bob generates a secret color **blue B**
- Alice and Bob agree on a common, public color **green G**
- They both mix their secret colors with **G**, so Alice has **green-amber GA**, and Bob has **green-blue GB**
- Alice sends **GA** to Bob, and Bob sends **GB** to Alice
 - Note: Eve now knows the colors **GA** and **GB**! Assume that it is hard to separate colors.
- Alice knows **GB**, so she can mix in **A** to form green-amber-blue **GAB**. Bob knows **GA**, so he can mix in **B** to form **GAB**, as well!
 - Eve only knows **G**, **GA**, and **GB**, so she can only form **green-amber-green-blue GAGB**, which is not the same!



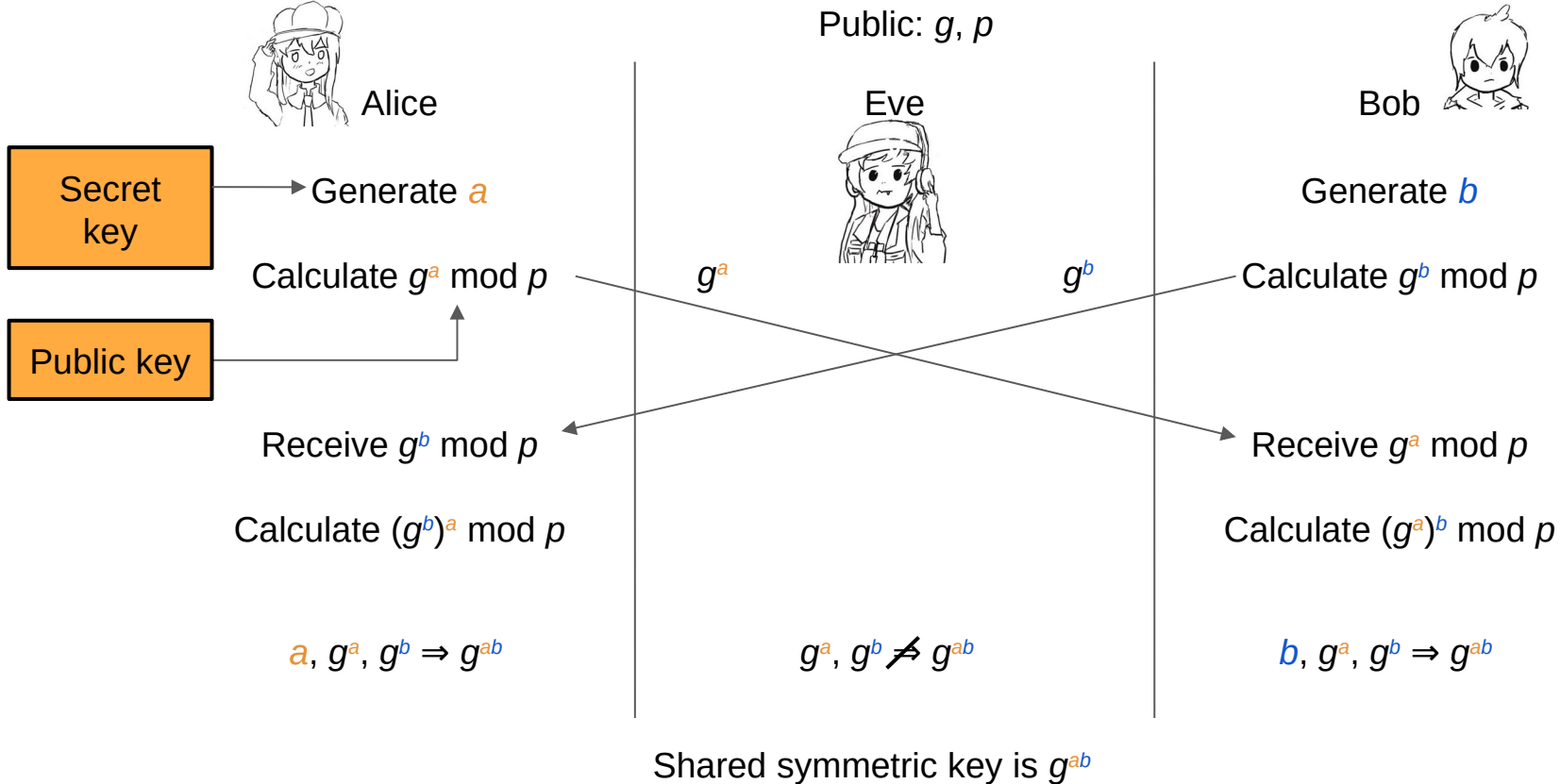
Discrete Log Problem and Diffie-Hellman Problem

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- There is a mathematical version of this
- Assume everyone knows a large prime p (e.g. 2048 bits long) and a number g
- **Discrete logarithm problem (discrete log problem):** Given $g, p, g^a \bmod p$ for random a , it is computationally hard to find a
- **Diffie-Hellman assumption:** Given $g, p, g^a \bmod p$, and $g^b \bmod p$ for random a, b , no polynomial time attacker can distinguish between a random value R and $g^{ab} \bmod p$.
 - Intuition: The best known algorithm is to first calculate a and then compute $(g^b)^a \bmod p$, but this requires solving the discrete log problem, which is hard!
 - Note: Multiplying the values doesn't work, since you get $g^{a+b} \bmod p \neq g^{ab} \bmod p$

Diffie-Hellman Key Exchange

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Discrete Log Problem and Diffie-Hellman Problem

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- g, p are publicly shared
- a, b are secret to Alice and Bob. Only Alice knows a , and only Bob knows b .
- $g^a \bmod p$, and $g^b \bmod p \Rightarrow$ these too are publicly shared.
- g is called the generator (also called primitive root mod p)
 - g lies within $[2, p-1]$
 - $g^a \bmod p$ distributes uniformly over $[1, p-1]$ for all possible choices of a .
 - In the context of number theory, if g is a primitive root of p , then $g^a \bmod p$ has residues $[1, p-1]$ and it has a period of $p-1$.
 - All possible choices of “ a ” includes: $[0, p-2]$. As soon as “ a ” becomes $p-1$, the modulo starts repeating from 1.
 - According to Euler’s theorem, $g^{\phi(p)} \equiv 1 \bmod p$. Since p is a prime number, $\phi(p) = p-1$.

Discrete Log Problem and Diffie-Hellman Problem

In Diffie–Hellman key exchange, the values a and b (private keys) must satisfy these properties:

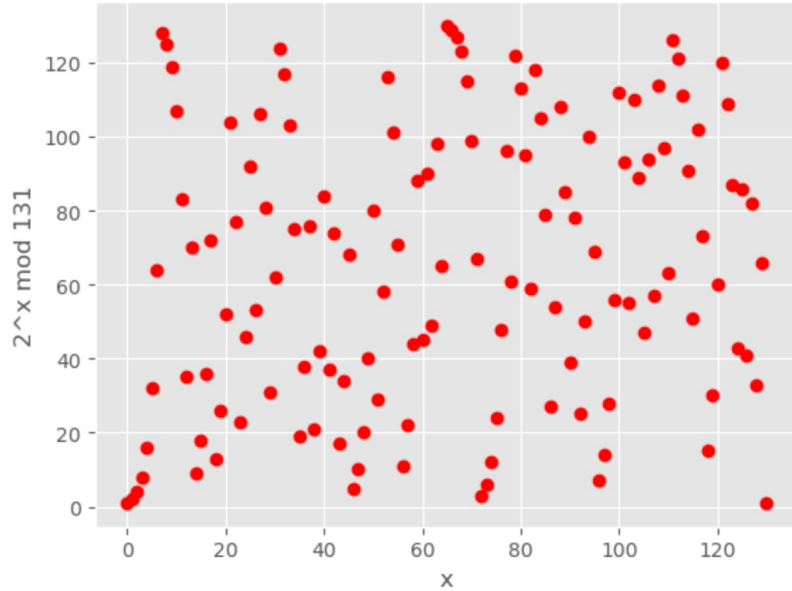
1. **Randomly chosen:** They should be chosen randomly and kept secret.
2. **In the range:** $1 \leq a, b \leq p - 2$, where p is a large prime.
3. **Relatively prime not required:** They do **not** need to be coprime to $p - 1$.
4. **Sufficiently large:** To ensure security, a and b should be large integers (hundreds or thousands of bits in real applications).

These private keys are then used to compute public keys:

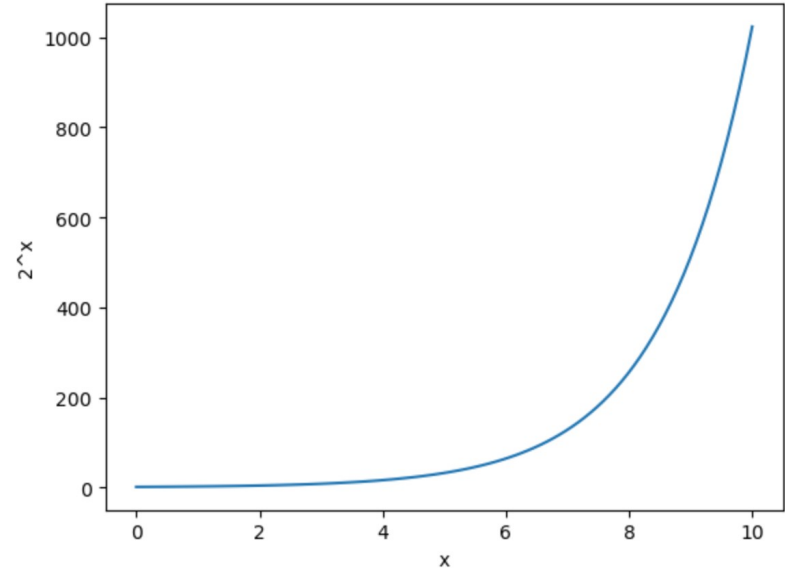
- $A = g^a \bmod p$
- $B = g^b \bmod p$

Discrete Log Problem

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Plot of $g^a \bmod p$



Plot of g^a

<https://www.desmos.com/calculator/3j0h11za2s>

Why must g be a primitive root?

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- If g isn't a generator (primitive root), then
 - $g^{ab} \bmod p$ only covers a subset of $[1, 2, \dots, p-1]$
 - The shared secret $g^{ab} \bmod p$ lives in that smaller subgroup, so it has less entropy
 - Less entropy = more predictable = easier to break / brute force.

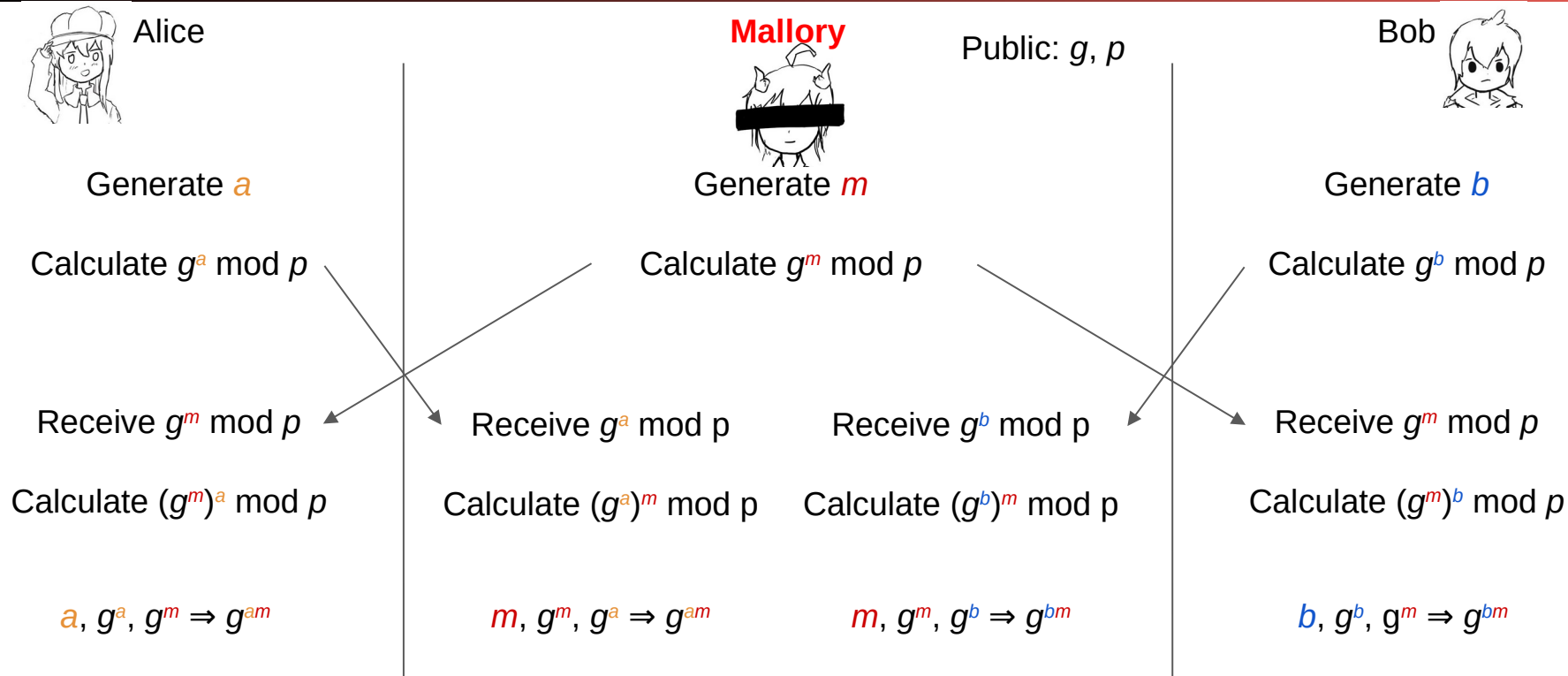
Ephemerality of Diffie-Hellman

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- Diffie-Hellman can be used ephemerally (called Diffie-Hellman ephemeral, or DHE)
 - **Ephemeral**: Short-term and temporary, not permanent
 - Alice picks a fresh random a , Bob picks a fresh random b for each session.
 - They derive their shared secret $K = g^{ab} \bmod p$
 - Once the session ends, they delete a , b and K .
 - Even if Eve steals Alice and Bob's secrets somehow later on, those values are going to be different, so Eve cannot decrypt any of the past messages.

Diffie-Hellman: Man In the Middle Attack

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Diffie-Hellman: Issues

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- Diffie-Hellman is not secure against a MITM adversary
- DHE is an *active protocol*: Alice and Bob need to be online at the same time to exchange keys
 - What if Bob wants to encrypt something and send it to Alice for her to read later?
 - Next time: How do we use *public-key encryption* to send encrypted messages when Alice and Bob don't share keys and aren't online at the same time?
- Diffie-Hellman does not provide *authentication*
 - You exchanged keys with someone, but Diffie-Hellman makes no guarantees about who you exchanged keys with; it could be Mallory!

Diffie-Hellman: Exercise

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(c) Explain the philosophy behind Diffie-Hellman key exchange protocol. In the Diffie-Hellman key exchange protocol, let the prime number be 353 and one of its prime root is 3. Let the users A and B select their secret keys 97 and 233, respectively. Compute the public keys of A and B and also their common secret key.

(3+7=10)

<https://www.khanacademy.org/computing/computer-science/cryptography/modarithmetic/a/fast-modular-exponentiation>

Group

A **group** $(G, \star)$ is a set G plus a binary operation $\star$ satisfying:

1. **Closure:** $a, b \in G \Rightarrow a \star b \in G$
2. **Associativity:** $(a \star b) \star c = a \star (b \star c)$
3. **Identity:** $\exists e \in G$ with $e \star a = a \star e = a$ for all a
4. **Inverses:** $\forall a \in G, \exists a^{-1} \in G$ with $a \star a^{-1} = a^{-1} \star a = e$.

Elliptic-Curve Diffie-Hellman (ECDH)

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- Notice: The discrete-log problem seems hard because exponentiating integers in modular arithmetic “wraps around”
 - Diffie-Hellman can be generalized to any mathematical group that has this cyclic property
 - Discrete-log uses the “multiplicative group of integers mod p under generator g ”
- Elliptic curves: A type of mathematical curve
 - Big idea: Repeatedly adding a point to itself on a curve is another cyclic group
 - You don’t need to understand the math behind elliptic curves
- **Elliptic-curve Diffie-Hellman**: A variation of Diffie-Hellman that uses elliptic curves instead of modular arithmetic
 - Based on the elliptic curve discrete log problem, the analog of the discrete log problem
 - Benefit of ECDH: The underlying problem is harder to solve, so we can use smaller keys (3072-bit DHE is about as secure as 384-bit ECDHE)

Summary: Diffie-Hellman Key Exchange

CSE 405

- Algorithm:
 - Alice chooses a and sends $g^a \bmod p$ to Bob
 - Bob chooses b and sends $g^b \bmod p$ to Alice
 - Their shared secret is $(g^a)^b = (g^b)^a = g^{ab} \bmod p$
- Diffie-Hellman provides forwards secrecy: Nothing is saved or can be recorded that can ever recover the key
- Diffie-Hellman can be performed over other mathematical groups, such as elliptic-curve Diffie-Hellman (ECDH)
- Issues
 - *Not* secure against MITM
 - Both parties must be online
 - Does not provide authenticity